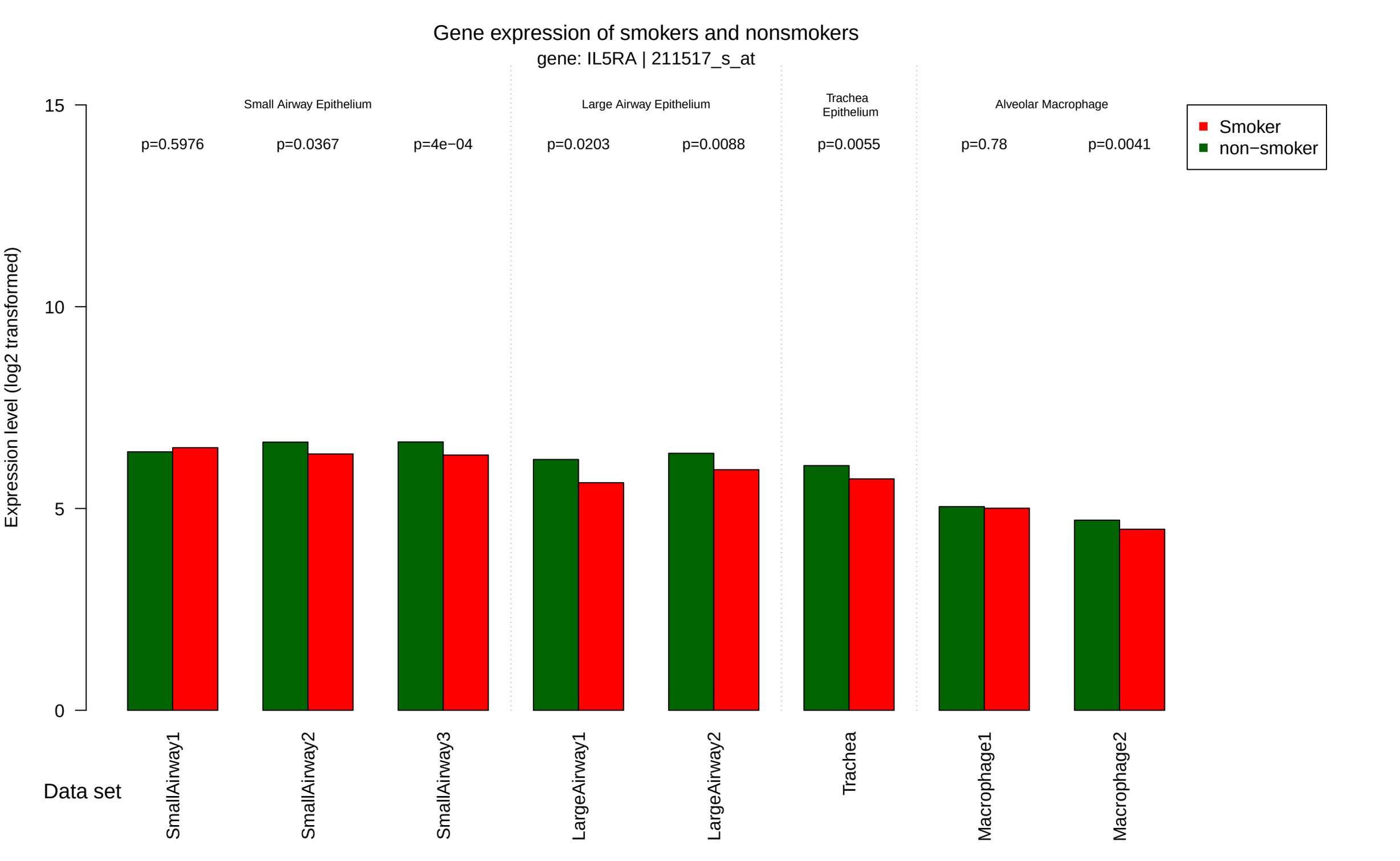
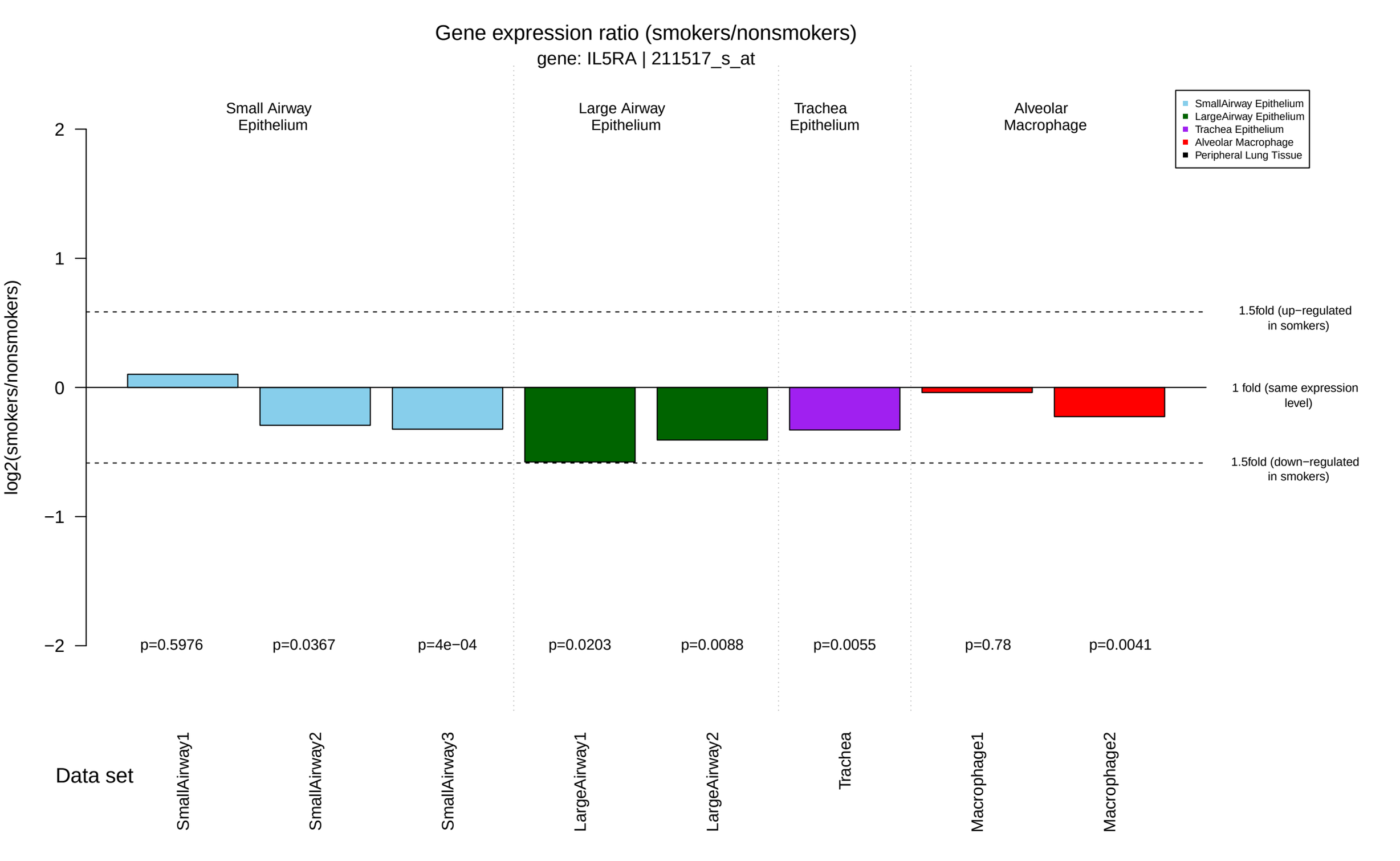
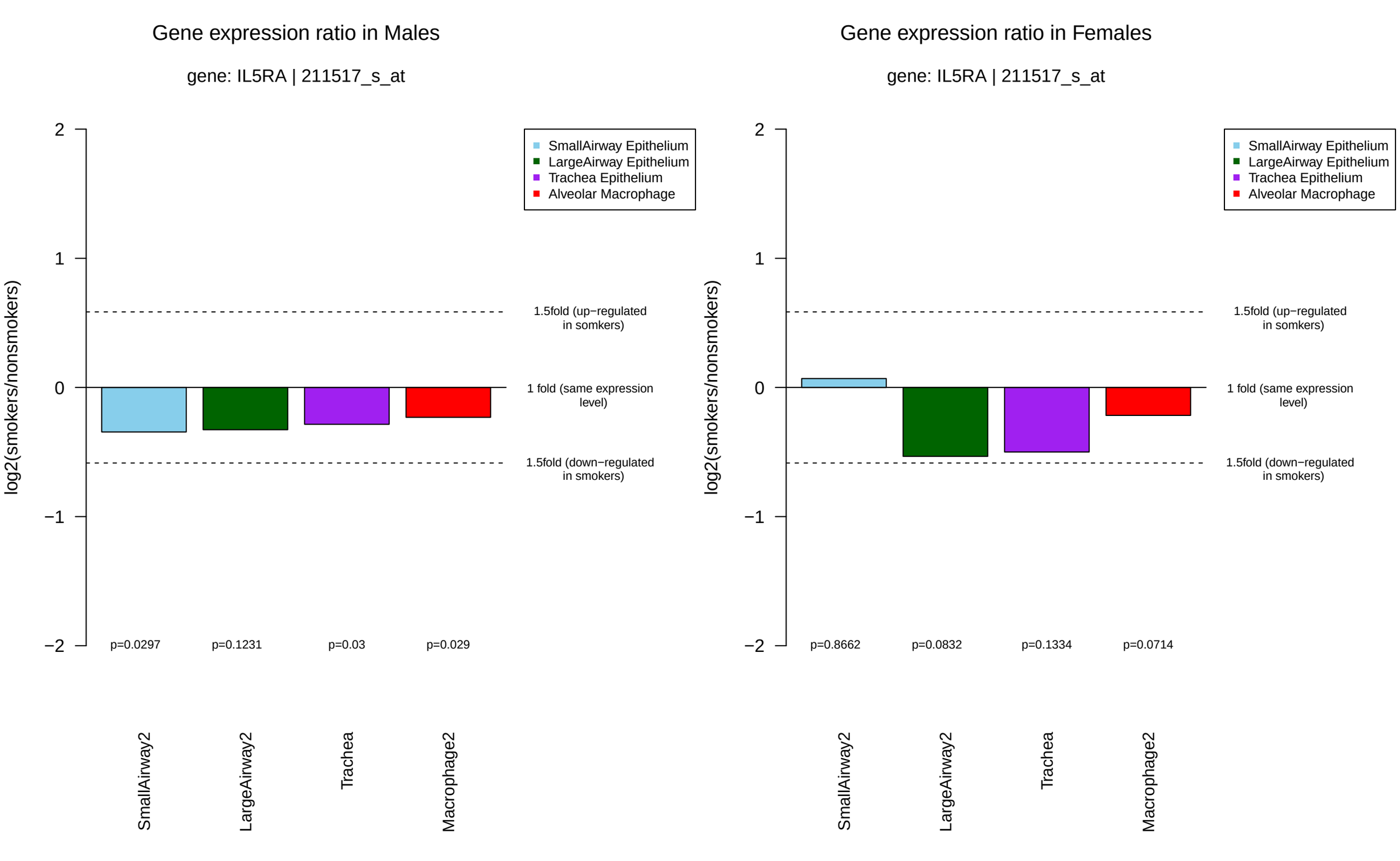
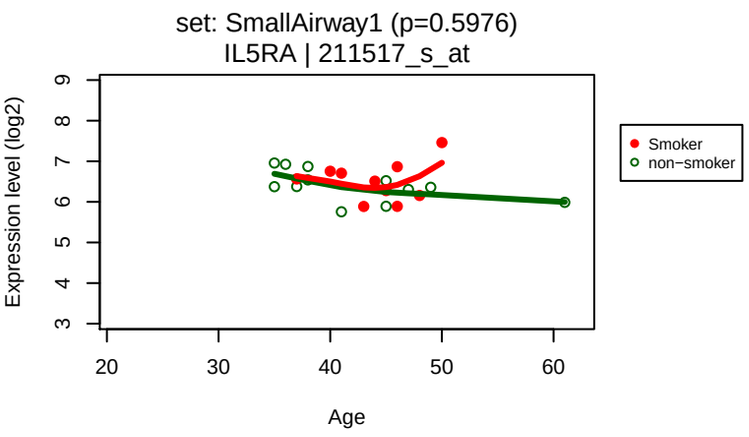
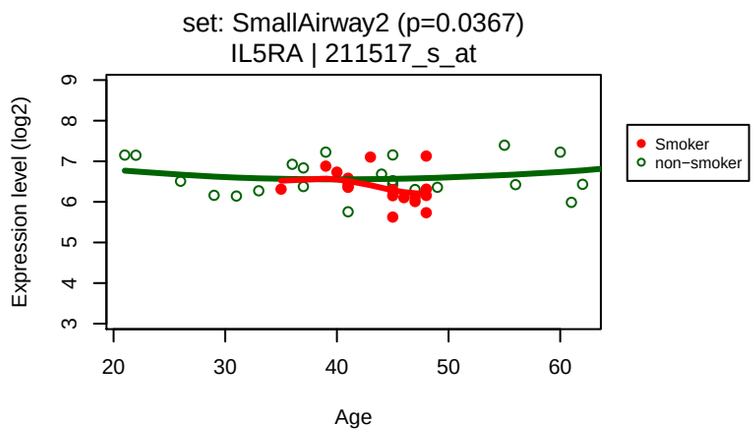
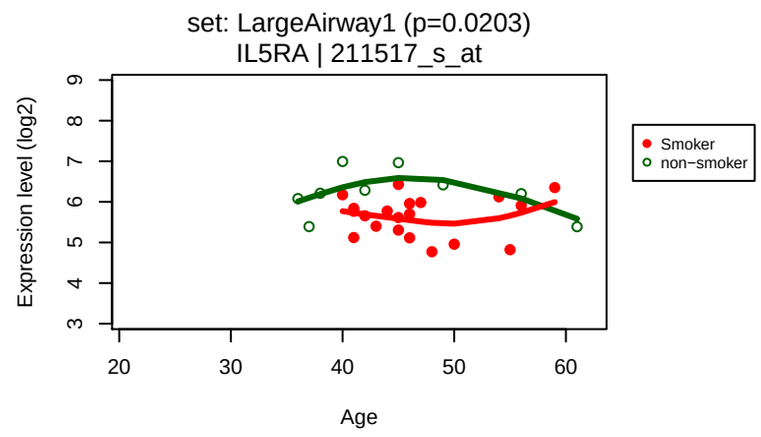

Gene expression of smokers and nonsmokers
gene: IL5RA | 211517_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5976
p=0.0367
p=4e−04
p=0.0203
p=0.0088
p=0.0055
p=0.78
p=0.0041
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

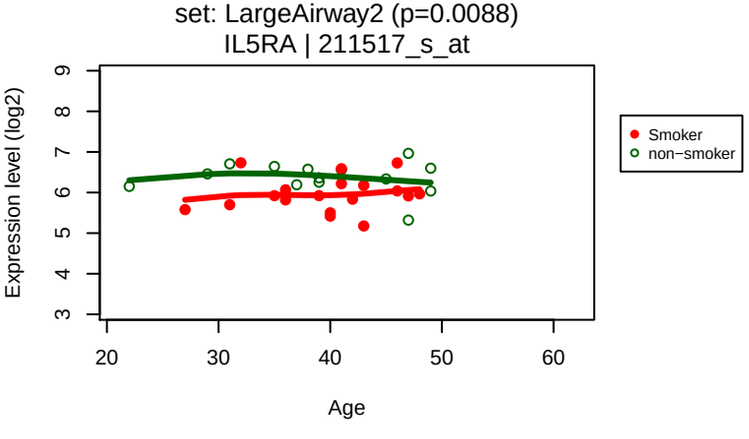

Gene expression ratio (smokers/nonsmokers)
gene: IL5RA | 211517_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5976
p=0.0367
p=4e−04
p=0.0203
p=0.0088
p=0.0055
p=0.78
p=0.0041
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

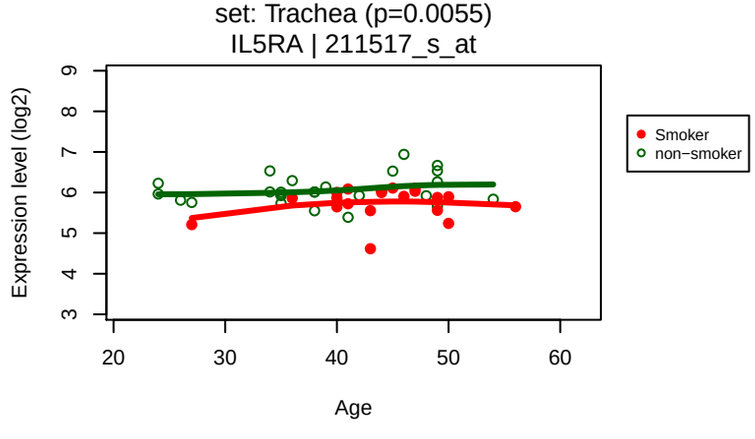

Gene expression ratio in Males
gene: IL5RA | 211517_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0297
p=0.1231
p=0.03
p=0.029
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL5RA | 211517_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8662
p=0.0832
p=0.1334
p=0.0714
SmallAirway2
LargeAirway2
Trachea
Macrophage2

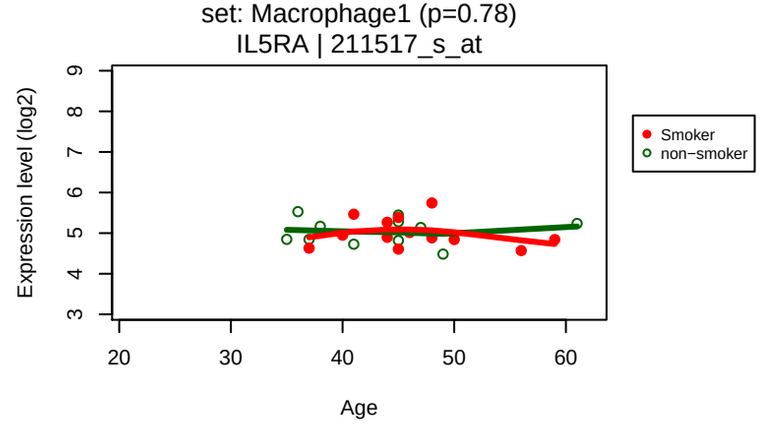

set: SmallAirway1 (p=0.5976)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker

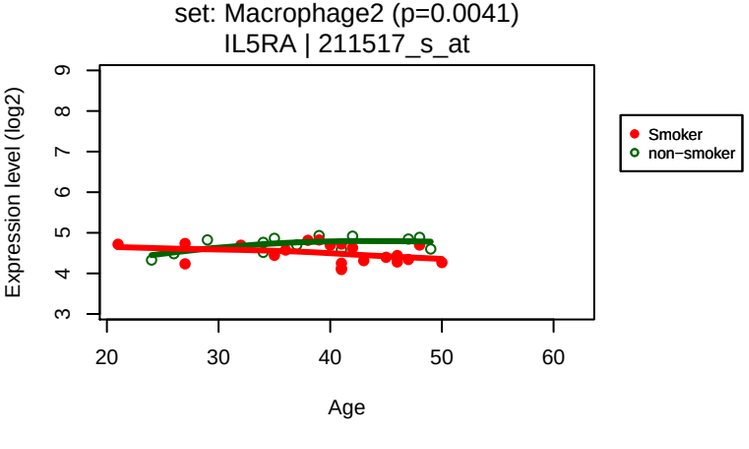

set: SmallAirway2 (p=0.0367)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker

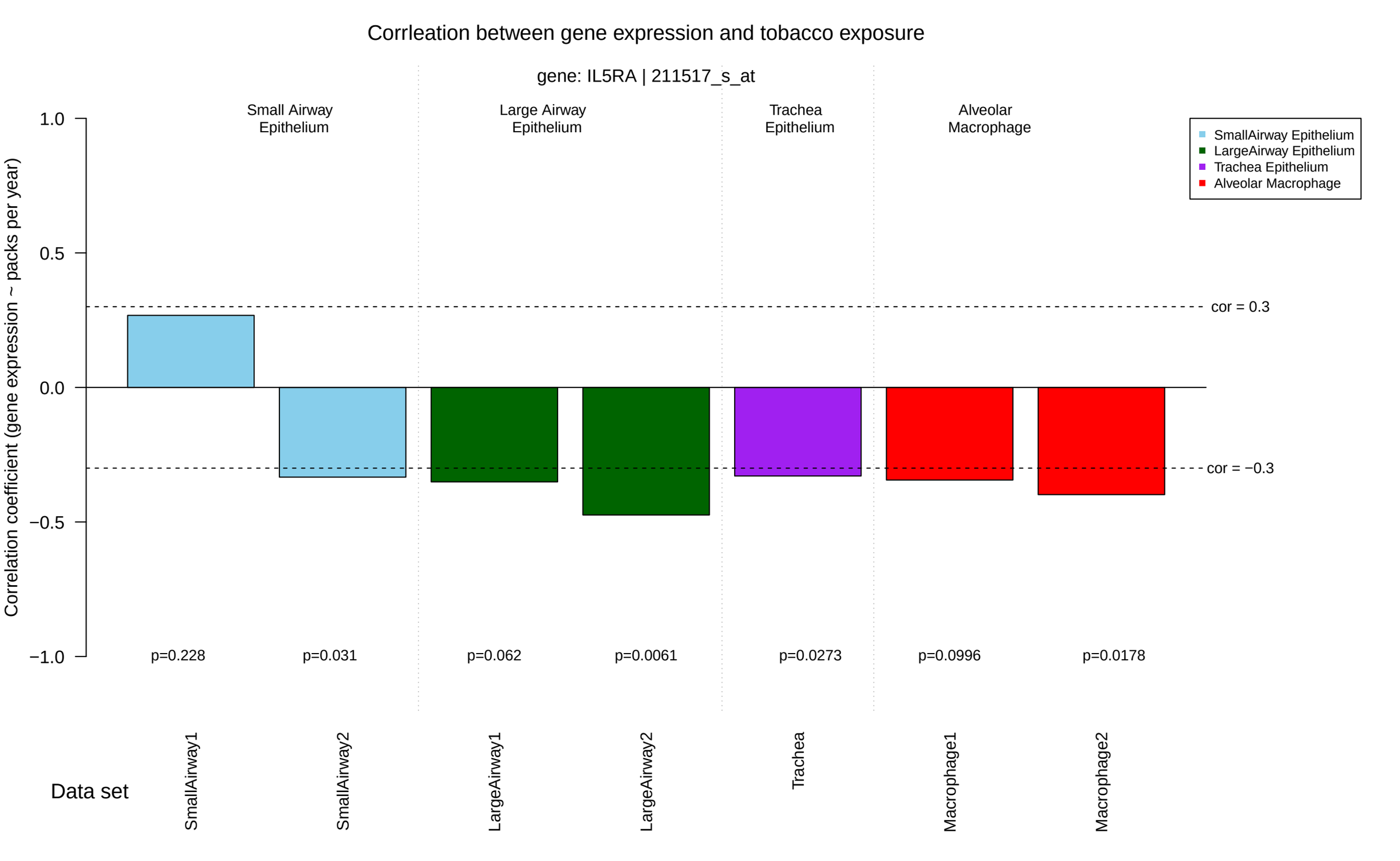

set: LargeAirway1 (p=0.0203)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0088)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0055)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.78)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0041)
IL5RA | 211517_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL5RA | 211517_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.228
p=0.031
p=0.062
p=0.0061
p=0.0273
p=0.0996
p=0.0178
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2